

Research Interests

Technical Reinforcement Learning, Stochastic and Linear Bandits, Distributed Optimization, Statistical Inference
Application Large Language Models (RAG and Agents), Efficient Federated Learning

Education

2022– **Cornell University**, *Doctorate of Philosophy (Ph.D.)*, 4.0/4.0.
Electrical and Computer Engineering Advisor: Prof. Vikram Krishnamurthy

2018–2022 **Indian Institute of Technology Guwahati**, *Bachelor of Technology*.
Major in Electronics and Communication Engineering GPA – 9.54/10 | 1st in batch of 128
Minor in Computer Science Engineering GPA – 9.4/10 | 1st in batch of 45

Experience

Jun 2025 – **PhD Research Intern**, MORGAN STANLEY, Machine Learning Research.

Aug 2025

- Developed a modification of the standard RL algorithm for LLMs, which performs a mixture of tokens generation to enable better exploration during the thinking phase
- Benchmarked performance of the modified techniques on 15+ environments from the reasoning gym and standard math benchmark to get modest gains of 5 – 40%, and upto 50% reduced memory requirements

May 2024 – **PhD Research Intern**, ADOBE RESEARCH, Digital Experience Cloud.

Aug 2024

- Developed an online algorithm for selecting samples to be annotated, which leverages difficulty feedback from the annotator and considers the constraint that only a single expert is available
- Proved regret guarantees for an explore and exploit algorithm for high dimensional sparse bandits and showed that it achieves sublinear regret even with a blocking constraint and without the hard sparsity constraint

Jul 2023 – **Graduate Assistant**, CORNELL CENTER FOR SOCIAL SCIENCES, Cassian D' Cunha.

May 2024

- Managed the cloud infrastructure for the CCSS, which provides computational resources for researchers
- Improved log analytics, preemptive measures, and resource scalability for the Azure-based server environment

May – Jul 2021 **Summer Analyst**, GOLDMAN SACHS, Cross Asset Quant Strats.

- Clustered counterparties using Frequent Itemseting for Credit Valuation Adjustment (CVA) calculations
- Improved computational performance by up to 40% for Foreign Exchange and Commodities CVA calculations
- Pre Placement Offer** was extended for a full-time role based on performance

May 2020 – **Research Assistant**, HAAS SCHOOL OF BUSINESS, UC BERKELEY, Prof. Abhishek Nagaraj.

Jul 2022

- Helped parameterize and program an experiment on the streetlight effect of information on exploration
- Analyzed and modeled heterogeneity in business's closure policies in response to Covid-19
- Created a [dashboard](#) for the impact of different reopening policies on health and economic outcomes

Publications

Journal **Interacting Large Language Model Agents. Interpretable models and social learning.**, A. Jain, V. Krishnamurthy, IEEE Access 2025, [Open Access Link](#).

Journal **Structured Reinforcement Learning for Incentived Stochastic Covert Optimization**, A. Jain, V. Krishnamurthy, Control System Letters (L-CSS and CDC), May 2024, [Code](#).

Journal **Controlling Federated Learning for Coverttness**, A. Jain, V. Krishnamurthy, Transactions on Machine Learning Research (TMLR), January 2024, [OpenReview](#), [Code](#).

Journal **Controlling stochastic gradient descent using stochastic approximation for robust distributed optimization**, A. Jain, V. Krishnamurthy, Numerical Algebra, Control and Optimization, 2025, [AIMS](#).

Journal **Interpretable Deep Image Classification using Rationally Inattentive Utility Maximization**, K. Pattanayak, V. Krishnamurthy, A. Jain, IEEE Journal of Selected Topics in Signal Processing, 2024.

In Review **Sparse Linear Bandits with Blocking Constraints**, A. Jain et. al., [arXiv](#).

- In Review **Stochastic Bandits with Mixture Reward Distribution**, A. Jain, Alec Koppel, Sujay Bhatt.
- Conference **Information Diffusion and Preferential Attachment in a Network of Large Language Models**, A. Jain et. al., Accepted, CDC 2025, [arXiv](#).
- Conference **Identifying Hate Speech Peddlers in Online Platforms. A Bayesian Social Learning Approach for Large Language Model Driven Decision-Makers**, A. Jain, V. Krishnamurthy, CDC 2024, [Code](#).
- Conference **Bimodal Bandits: Max-Mean Regret Minimization**, A. Jain, S. Bhatt, V. Krishnamurthy, A. Koppel, Asilomar Conference, July 2024, Accepted, *Finalist for Top Student Paper Award*.
- Conference **Optimal Joint Antenna Selection and Beamforming for an Intelligent Reflecting Surfaces Aided Multiuser System**, A. Jain, S. Kashyap, IEEE WCNC, Dec 2023, [IEEE Link](#).
- Conference **Low Complexity Passive Beamforming Algorithms for Intelligent Reflecting Surfaces with Discrete Phase-Shifts over OFDM Systems**, A. Jain, R. Gowda, S. Kashyap, R. Sarvendranath, National Conference on Communications, May 2022, Accepted, [IEEE Link](#).

Technical Skills

- Languages Python, R, MATLAB, C++, JavaScript
- Frameworks PyTorch, PySpark, Pandas, axolotl, langchain, Plotly, numpy, OpenCV
- Web Tech. jQuery, d3.js, React, Django, Flask, HTML, CSS
- Presentation $\LaTeX$, Figma, Powerpoint

Achievements & Honours

- 2023-2025 **Data Science Fellowship**, Cornell Center for Social Sciences.
- 2022 **Institute Silver Medalist**, IIT Guwahati, Graduated top of the class in a batch of 144 students.
- 2020 - 21 **Institute Merit Scholarship 2021**, IIT Guwahati, full tuition fee waiver for ranking 1st in department.
- 2019 - 20 **Institute Merit Scholarship 2020**, IIT Guwahati, full tuition fee waiver for ranking 1st in department.
- 2018 **JEE Advanced**, Secured 99.996 percentile among 150K students with a rank of 1117.
- 2018 **JEE Mains**, Secured 99.999 percentile among 1.5M students with a rank of 237.

Relevant Courses * AS/Outstanding Grade

- Math. & EECS Stat. Learning Theory, Algorithmic Game Theory, Probability, Mathematical Statistics, Bayesian Estimation and Stochastic Optimization*, Statistical Algorithms*, Data Structures & Algorithms*

Additional Coursework

- Math Probability & Random Processes, Linear Algebra, Multi-variable Calculus, ODEs
- ECE Network Coding, Information Theory & Coding, Digital Circuits*, Video Analytics*, Digital Communications*, Digital Signal Processing, Data-Driven System Theory, Adv. Control Systems

References

Vikram Krishnamurthy , <i>Professor</i> , ECE, Cornell University, vikramk@cornell.edu. <i>Advisor</i>	Brendan Hogan , <i>Research Scientist</i> , MLR, Morgan Stanley, bhogan.net. <i>Internship Mentor</i>	Soumyabrata Pal , <i>Research Scientist</i> , Adobe Research, soumyabratap.github.io. <i>Internship Mentor</i>
Salil Kashyap , <i>Assistant Professor</i> , EEE, IIT Guwahati, salilkashyap@iitg.ac.in. <i>BTP Supervisor</i>	Abhishek Nagaraj , <i>Assistant Professor</i> , Haas UC Berkeley, nagaraj@berkeley.edu. <i>RA Supervisor</i>	